# A Universal and Reconfigurable Stability Control Methodology for Articulated Vehicles With Any Configurations

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***Abstract*—To meet with many different transportation needs, it comes in a rich diversity and variety of articulated vehicles. Vehicle combinations are seen in different axle configurations, number of articulations, powertrain, active actuation systems, etc. This research is, therefore, motivated to develop a model-based control system in a universal and reconfigurable fashion to any articulated vehicles stability control. To achieve its universality and reconfigurability, we introduce a hierarchical (two-layer) control system. Namely, the high layer formulates a model predictive control (MPC) tracking problem to generate corrective Center of Gravity (C.G.) forces/moment. The lower-level controller is formulated as Control Allocation (CA) algorithm to regulate steering or torque (brake) at each wheel optimally and reconfigurable as to meet high-level calculations. Real-time constraints, i.e. actuator limits, tire capacity, and actuator failure are discussed. Diverse applications are presented that the universal and reconfigurable methodology is handy, capable and effective on stability control while applying to various vehicle configurations and objectives.**

***Index Terms*—Articulated vehicles, control allocation, integrated vehicle stability, model predictive control, universal and reconfigurable control.**

## I. Introduction

IN ORDER to transport a larger volume of freights or passengers via a relatively fixed infrastructure, one way to increase transportation efficiency is to employ longer vehicles with more axles or multiple articulated units [1], [2]. These vehicles are used most often in trucks, tractor-trailers, coaches/uses, and less frequently on passenger cars.

### A. Background

Articulated multi-axle vehicles, e.g., trucks, are the main transportation means of goods all around the world. More than 80% of inland freight volume is transported by road vehicles today. Each year, road transport carries 6,000 billion tonne-kilometres of goods in the EU, USA, CIS, China, and Japan alone. In the concept of 'bigger is better,' the bigger the vehicle within a practical limitation, the more payload that can be transported with just one driver. In a public transport system, buses have been playing a significant role with their advantageous features in carrying capacity, transport productivity and reduction of transport costs [3]. Buses have more than two axles usually three (known as a tri-axle bus). Extra axles are usually added for legal weight restriction reasons, or usually we can see using articulation to implement trailer buses.

The popularity of such vehicles, however, garnered increased public attention to safety problems. Annually, approximately 500,000 accidents involving trucks or buses occur in U.S. Of the approximately 415,000 crashes involving large trucks in 2016, there were 4440 fatal crashes and estimated 119,000 injury crashes [4], [5]. Worse yet, the majority of accidents with heavy trucks involved are trucks against passenger cars, where the severity is much worse for the passenger car due to its vulnerability. Nevertheless, nowadays, only a minority of these commercial vehicles are equipped with stability control systems.

### B. Diversity, Challenges and Attempts

Articulated vehicles differ from the passenger car quite a lot and it makes a big challenge for dynamics controls. One of them is to design a controller that applies to diverse configurations of these vehicles. Distinctive configurations might be seen in several dimensions. 1) Axle/articulation configuration. To meet many different transportation purposes, a wide range of combinations exists. There are vehicles with one unit of multiple axles or multiple articulated units where each unit could be different. 2) Powertrain/steering configuration. The powertrain of a vehicle describes the main components that in which way it generates power and how it delivers to the road surface. Depending on the needs, there are various drive modes and steer modes, seen in [6]. 3) Active actuators configuration. Active actuators determine the active control strategy in lower level implementation. Common actuators are: a) active torque differential, including torque vectoring and differential braking [7], [8]. b) active steering systems [9]–[11]; c) (semi)active suspension systems, including anti-roll bar [12]–[14]; Considering attributes such as physical complexity and cost, active actuators configurations vary from vehicle to vehicle.

Another challenge comes from the complex dynamics characteristics. Multi-axle vehicles are usually characterized by a high wheelbase to track ratio, a high center of gravity (CG), and uneven weight distribution between axles, prone to instability or

Manuscript received February 26, 2019; revised November 1, 2019; accepted February 5, 2020. Date of publication February 11, 2020; date of current version April 16, 2020. The review of this article was coordinated by Dr. A. Heydari.
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Digital Object Identifier 10.1109/TVT.2020.2973082

even roll over. Especially, the existence of articulation makes it complex while developing the dynamics modeling and control system. Compared to passenger cars, more control objectives come up, for instance, trailer sway, jackknife, off-tracking, rearward amplification. These objectives have to be prioritized considering the vehicle characteristics and application context. An ideal controller should be inclusive yet reconfigurable and selective to these stability objectives.

Although most work focus on two-axle vehicles, the state of the art for control strategies/concepts in literature inspire us go further. In [15], it well reviewed Integrated Vehicle Dynamics Control (IVDC) architectures, that are differentiated as centralized, supervisory, hierarchical, and coordinated control. Among them, research in integrated vehicle dynamics control( [8], [16]), and reconfigurable control using optimal control techniques ([17], [18]) are very active. For instance, [16] proposed unified chassis control(UCC) strategy for improving agility, maneuverability, and vehicle lateral stability by the integration of active steering and differential braking. A coordinated and reconfigurable vehicle dynamics control system in [17] is presented by high-level sliding mode control to achieve generalize forces, and distributed to the slip and slip angle of each tire by control allocation (CA)in a lower-level controller. In [18], a Holistic Corner Control (HCC) strategy is introduced, in which provides real-time vehicle stability control while minimizing tire forces and control efforts by formulating a quadratic programming optimization. More extended works in in [19] proposed a reconfigurable modelling and control approach that applied to an SUV with different types and combinations of control actuation.

### C. Contributions and Novelties

Under the challenges of configuration diversity, multiple objectives, and actuation redundancy in articulated vehicles, the main contribution as well as the core goal of this paper is to develop a universal and reconfigurable methodology that could apply to any configurations of these vehicles for dynamics control. The reconfigurable modeling and control framework has been introduced and effectively applied to any multi-axle (non-articulated) vehicles, reported in [20], [21], nevertheless articulated vehicles are excluded. Differentiating from the previous work, this framework introduced an extended modeling process, a newly designed controller and very different vehicle cases study in application part. Leveraging this methodology, there is no need to reformulate the control problem when a new vehicle with new configurations comes. One could just set the defined 'Boolean Matrix' values based on vehicle configuration and do the minor tuning, a new and ad-hoc controller will be formulated. It is hoped that this methodology will provide a new perspective on articulated vehicle dynamics control and aid us all in implementing many different active systems.

The paper is structured as follows. Section II briefly describes the framework of reconfigurable modeling which serves for model-based controller design. Section III outlines the potential control objectives in yaw and roll planes. Section IV formulates the reconfigurable controller from prediction model to high-level and lower-level schemes, and discusses real-time constraints. Section V comes to four cases with different configurations of axle, powertrain, active actuator and different emphasis on control purposes. Section VI concludes the paper along with recommendations and future work.

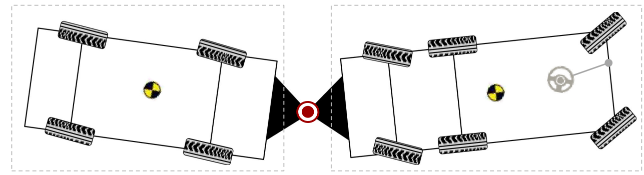


Fig. 1. From a single unit to articulated multi-unit vehicle.

## II. Dynamics and Control-Oriented Modeling

### A. Philosophy of Reconfigurable Modeling

The reconfigurable modeling to any vehicles is originally elaborated in [20], where details, e.g. derivation process and symbols/nomenclature are well explained. It should be mentioned that vehicle unit configured with 2 to 4 axles are only reported there for the sake of demonstrating simplicity. It finalizes 'One Formulation' that units any multi-axle vehicles, no matter it is articulated or not. For the convenience of developing the controller, the modeling process is briefly introduced here.

In the modeling process, the vehicle dynamics is firstly derived from the tire forces to corner forces. Next, all the forces of the tires are equalized to the CG of the vehicle, which will result in CG forces/moment. At last, treating the body as a point mass, the desired dynamics equations are formulated. We novelly defined '*Actuator Boolean Matrix*' to determine the availability (or configuration) of the active actuation and '*Axle Boolean Matrix*,' the availability (or configuration) of axles, namely, the axle's quantity. How are articulated multi-unit vehicles then modeled? The benefits of 'reconfigurable modeling process of a single vehicle unit' is taken. In Fig. 1, the vehicle is articulated with two arbitrary units by a physical articulation. Following this intuition, the model can be formulated as a two vehicle units dynamics modeling developed, along with corresponding dynamics/kinematic constraints at articulation point. The '*Articulation Boolean Matrix*' is defined to judge the availability of the articulation, namely, the vehicle is articulated or not.

### B. A Single Unit Vehicle

To establish the modeling formulation of articulated vehicles, we start from the reconfigurable modeling of a single unit vehicle. Neglecting the effect of camber angles, it is assumed the vehicle equips with two basic actuators at each corner, which are driving/braking and steering. The relationship of corner forces and local(tire) forces including the actuator configuration is expressed as matrix form,

$$F_c = L_w \left( f + T_w \Delta f \right), \in \mathbb{R}^{16 \times 1} \tag{1}$$

where we defined corner force vector $F_c$ is the result of the normalized tire force vector $f/\Delta f$ with respect to $x$ and $y$ axis.

Vector $f$ includes each tire's longitudianal and lateral force while $\Delta f$ suggests the augmenting forces on the local forces applied by the active controller. $L_w$ is the mapping matrix. $T_w$ is defined as the '*Actuator Boolean Matrix*,' where configures the active actuators. Concretely, this matrix is changeable based upon what actuation system the vehicle has, for example, differential braking or active steering [20].

To map each wheel's corner forces to CG location, the total longitudinal force, total lateral force and total moment at CG are expressed as,

$$F_{CG} = L_c T_c F_c, \in R^{3\times 1} \tag{2}$$

where $L_c$ is the mapping matrix from corner forces to CG forces. As a result, the force vector at CG (x-y plane) is defined that $F_{CG} = [F_x \quad F_y \quad M_z]^{\mathrm{T}}$, wherein $F_x$ is the total CG longitudinal force; $F_y$, the total CG lateral force; $M_z$, the total CG yaw moment;. $T_c$ is defined as the '*Axle Boolean Matrix*,' which determines the availability of axles, i.e. how many axles the vehicle has.

Coupling 4 degree-of-freedom, i.e., the longitudinal, lateral, yaw, and roll motion, the linearized vehicle motion can be described as a state space form with the input of CG forces [20],

$$\dot{x} = Ax + BF_{CG} \tag{3}$$

where the state $x = [v_x \quad v_y \quad r \quad \phi \quad \dot{\phi}]^{\mathrm{T}}$ that represents the longtidunal speed ($v_x$), lateral speed ($v_y$), yaw rate($r$), and roll angle/ roll rate ($\phi/\dot{\phi}$).

In this research, the brush tire model is used to calculate the lateral tire force. For the computation's sake, it is linearized at the slip angle operating point considering the vertical load and road condition [22], [23]. Given a nonlinear explicit tire model, we take the zeroth and first-order terms of its Taylor expansion to achieve an affine model. The longitudinal tire force is proportional to the wheel torque assuming small slip ratios. From the single-track model, the tire slip angles can be expressed as vehicle state $x$ and steering angle. Given vehicle state $x$, driver input vector $w$, i.e torque and steering inputs, and active controller input $u$, we write the linearized tire (local) forces in matrix form,

$$f = A^{\mathrm{tire}} x + B^{\mathrm{tire}} w + d^{\mathrm{tire}} \tag{4}$$

$$\Delta f = B^{\mathrm{tire}} u \tag{5}$$

where the inputs vector are defined as:

$$w = \begin{bmatrix} w_1^{\mathrm{T}} & w_2^{\mathrm{T}} & \cdots & w_8^{\mathrm{T}} \end{bmatrix}^{\mathrm{T}}, w_i = \begin{bmatrix} Q_i & \delta_i \end{bmatrix}^{\mathrm{T}};$$

$$u = \begin{bmatrix} u_1^{\mathrm{T}} & u_2^{\mathrm{T}} & \cdots & u_8^{\mathrm{T}} \end{bmatrix}^{\mathrm{T}}, u_i = \begin{bmatrix} \Delta Q_i & \Delta\delta_i \end{bmatrix}^{\mathrm{T}}.$$

Symbol $Q/\delta$ denotes torque (driving or braking) and steering angle.

Combining the derived formulation of tires forces and vehicle dynamics in (1)-(5), the whole vehicle model yields the following expression as a standard state-space form in a continuous time,

$$\dot{x} = A_c x + B_c v + G_c w + d_c \tag{6}$$

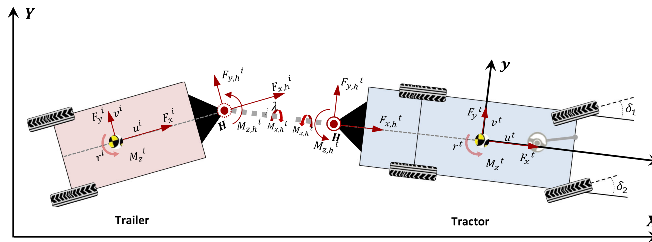


Fig. 2. Separated units of an articulated vehicle.

where $v = \begin{bmatrix} \Delta F_x & \Delta F_y & \Delta M_z \end{bmatrix}^{\mathrm{T}}$ is the virtual corrective CG forces to be calculated from the high-level controller. We name (6) as the 'reconfigurable vehicle model.' Given any multi-axle vehicles (one-unit), there is no need to start over again to model for every vehicle. It is readily achieved by easily configuring the 'Axle Boolean Matrix $T_c$' and 'Actuator Boolean Matrix $T_w$.'

### *C. Articulated Multi-Unit Vehicles*

Articulated vehicles dynamics is complex due to the coupled constraints at articulation. Many tried to model it directly through differential equations and complex transformations and items elimination. Our idea is simple and straightforward. Intuitively, the dynamics of articulated vehicles(one articulation) is 'dynamics of two separate units + constraints at hitch point.' Following the idea, now the modeling is moving from a single vehicle unit to articulated vehicle. Taking the tractor-trailer in Fig. 2 as an example, the articulated vehicle has 6 DoFs, where the tractor unit has the freedom to go forward, side-slip, yaw, and roll, while the trailer has the freedom to yaw relative to the tractor and to roll.

*Tractor/Trailer Dynamics:* Following the same method of linearization on one unit dynamics formulation (3), we separate two units and express as,

$$\dot{x}^n = A^n x^n + B^n F_{CG}^n + C^n F_h^n; (n = t, i) \tag{7}$$

where parameters or variables with superscript $t$ denotes to the tractor, and $i$ to the trailer. The only difference shown here is the force term $C^n F_h^n$ due to the existence of the articulation, $F_h^t/F_h^i$ denotes the hitch force vector with respect to the tractor/trailer while decoupling two units.

*Articulation Constraints:* Two groups of constraints are suggested at articulation point. The kinematic constraint, namely, the velocities $(v_x^i, v_y^i)$ at the CG of the trailer can be written with respect to the velocities at the CG of the tractor $(v_x^t, v_y^t)$ and the articulation angle ($\lambda$),

$$v_x^i = v_x^t \cos\lambda + (v^t - l_h^t r^t)\sin\lambda \tag{8}$$

$$v_y^i = -v_x^t \sin\lambda + (v_y^t - l_h^t r^t)\cos\lambda - l_h^i r^i \tag{9}$$

where $l_h^t/l_h^i$ is the distance between the articulation point and the CG of the tractor/trailer.

The dynamics constraint, namely, the relationship of the longitudinal and lateral force between the tractor and trailer at the articulation is written as,

$$F_{x,h}^t = -F_{x,h}^i \cos\lambda - F_{y,h}^i \sin\lambda \tag{10}$$

TABLE I
THE RULE OF ARTICULATED VEHICLES MODEL

| $T_\varepsilon$ | Rule Description |
|---|---|
| $(I_{10\times10})$ | Articulated vehicle, articulation constraints should be included |
| $(0_{10\times10})$ | No articulation exists, considered as separated units |

$$F^t_{y,h} = F^i_{x,h}\sin\lambda - F^i_{y,h}\cos\lambda \tag{11}$$

And the interaction yaw moment ($M^t_{z,h}$) and roll moment ($M^t_{x,h}$) at articulation yield,

$$M^t_{z,h} = -M^i_{z,h} = K_\lambda\lambda + C_\lambda\dot{\lambda} \tag{12}$$

$$M^t_{x,h} = -M^i_{x,h} = K_{t,i}(\phi^t - \phi^i) \tag{13}$$

where $K^{t,i}_{\lambda,h}$ and $C^{t,i}_{\lambda,h}$ are the articulation angle stiffness coefficient and damping coefficient,and $K^{t,i}_{\phi,h}$ is the roll stiffness of articulation point to calculate the roll moment due to the roll angle difference between two units.

*Model Finalization:* The constraints of the moment at articulation point and kinematic relationships are as demonstrated from (8) to (13). By summarizing above equations, the matrix form with the equality constraints of the articulated vehicle is written as,

$$\underset{\mathbf{2\,units}}{\text{s.s form.}} \quad \dot{x} = Ax + BF_{CG} + T_\varepsilon(CF_H)$$

$$\text{s. t.} \quad \begin{aligned} &F_H^t = L_{t-i}{F_H}^i \\ &v^i_x = v^t_x\cos\lambda + (v^t - l^t_h r^t)\sin\lambda \\ &v^i_y = -v^t_x\sin\lambda + (v^t_y - l^t_h r^t)\cos\lambda - l^i_h r^i \\ &\dot{\lambda} = r^i - r^t \end{aligned} \tag{14}$$

where,

$$x = \begin{bmatrix} x^t \\ x^i \end{bmatrix}, A = \begin{bmatrix} A^t & 0_{(5\times5)} \\ 0_{(5\times5)} & A^i \end{bmatrix}, B = \begin{bmatrix} B^t & 0_{(5\times5)} \\ 0_{(5\times5)} & B^i \end{bmatrix},$$

$$F_{CG} = \begin{bmatrix} {F_{CG}}^t \\ {F_{CG}}^i \end{bmatrix}, C = \begin{bmatrix} C^t & 0_{(5\times3)} \\ 0_{(5\times3)} & C^t \end{bmatrix}, F_H = \begin{bmatrix} {F_H}^t \\ {F_H}^i \end{bmatrix},$$

$T_\varepsilon$ is defined as '*Articulation Boolean Matrix*' in Table I to determine the availability of the articulation, i.e whether the vehicle is articulated or not. Note that, by following the modeling approach, we could readily extend the one-articulation vehicle to any number of articulations vehicles. Each unit of the vehicle is allowed to allocate any axles, and each axle is allowed to steer or drive/brake independently. Readers can find the work of extension to any articulated $n$-unit vehicles in [20].

## III. Stability Control Objectives

Vehicle stability references (or desired states) are well studied from vehicle fundamental dynamics, which are not repeated here. As the control-oriented model includes 4 DoF of each unit, the desired vehicle response consists of the desired longitudinal speed, desired yaw rate, desired lateral speed, and desired rollover index. Readers can find the detailed derivation in listed literature. Although not studied in this paper, longitudinal dynamics is included in the modelling. Desired longitudinal speed is set by the driver or active controller, for instance, in adaptive cruise control/emergency braking control. Vehicle lateral speed (or slip angle) is unavoidable in cornering. The controller only articulates the lateral speed while large body slip occurs [24]. Desired lateral speed ($v_{yd}$) is hence defined as a envelop control, which reduces frequent control activation and energy consumption. The desired yaw rate ($r_d$) is obtained from interpreting the driver's intention. [20] reports how to derive the desired yaw rate for multi-axle vehicles. Rollover index ($RI$) is a common coefficient that reflects the vehicle lateral load transfer. In this research, un-tripped rollover is considered and a general $RI$ is used for vehicle rollover control [25]. Alternatively, [26], [27] proposed the zero-moment point (ZMP) based on a simplified kinematic model of the vehicle while conventional approaches require tyre forces measurement/estimation. The ZMP is analyzed and validated across several vehicle configurations and this would be beneficial when limited sensors are equipped in the vehicle. Similar to lateral stability, the desired rollover index ($RI_d$) is only tracked when exceeding a defined threshold ($RI_{\max}$) [28].

Generally, for articulated 2-unit vehicles (tractor-trailer), we control the articulation angle instead of the trailer yaw rate, to prevent instability such as trailer sway and jackknife. The desired articulation angle is developed by considering the steady-state handling behavior [29]. By referring to the articulation angle gain with respect to the front steer angle, it is achieved as,

$$\lambda_d = \frac{(L^i + l^t_h) + K^i_{us}v^2}{L^t + K^t_{us}v^2}\delta \tag{15}$$

where $K^t_{us}$ is the understeer coefficient of the tractor unit while $K^i_{us}$, that of the trailer unit. $L^t/L^i$ is tractor/trailer wheelbase. To make sure both units are directional stable, both units are designed to be understeer. $K^t_{us}$ and $K^i_{us}$ are designed to be very small positive. They are calculated by the axle cornering stiffness and its weight distribution.

## IV. Universal Controller Design

To achieve the universality, reconfigurability, and integratability of the controller, a two-layer control structure was developed, illustrated in Fig. 3, It could apply to any multi-axle vehicles with any control objectives in yaw and roll dynamics plane. However, depending on a specific application, part of the CG corrective forces, e.g. only the CG corrective yaw moment, and selected objectives may be adopted. The objective of control strategy falls into two layers: At each time step, the high-level controller is to ensure the vehicle follows the state references, and guarantees stability and safety by solving the following MPC problem:

$$\begin{aligned} \min \quad & \text{tracking error} + \text{CG corrections} + \text{oscillation} \\ & \text{vehicle dynamics model} \\ \text{s.t.} \quad & \text{input boundary} \\ & \text{feasibility and stability} \end{aligned}$$

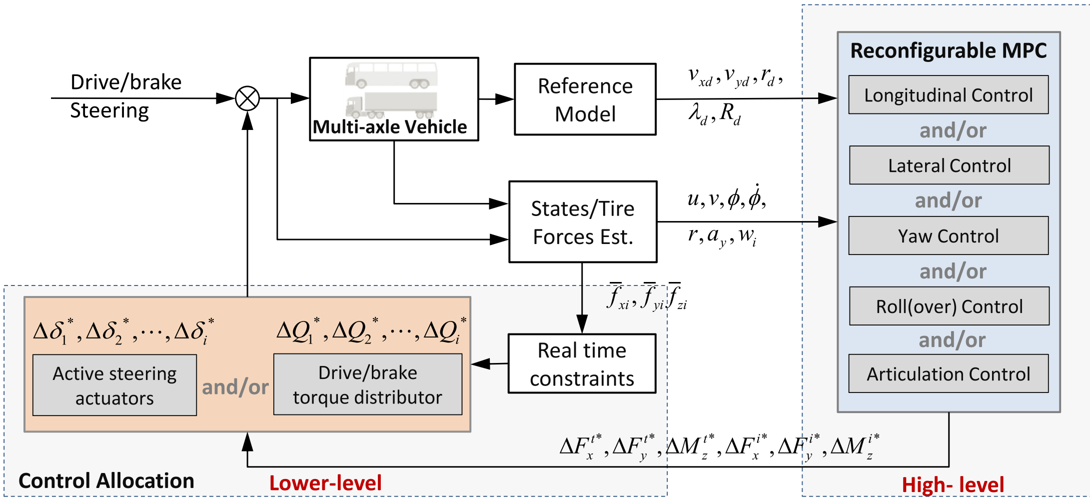


Fig. 3. A complete structure of controller.

Once the optimal CG corrections are computed, lower level controller allocates the wheel steering, drive/braking torque optimal by solving the following control allocation problem:

$$\begin{aligned} &\min \quad \text{mapping error} + \text{control efforts} \\ &\text{s.t.} \quad \text{input boundary} \end{aligned}$$

Basically, the proposed controller framework consists of four parts, namely, the reference model module, high-level controller, lower-level controller, and real-time constraints, whereas the estimation module is out of the paper's scope. The driver's steering and drive/brake torque commands are passed to the vehicle. Based on that, the desired dynamics states ($y_d$) are calculated together with other information. The high-level controller uses a MPC algorithm to optimal corrective CG forces ($\Delta F_x$, $\Delta F_y$, $\Delta M_z$) of each unit by minimizing the errors between the reference sequences and sequences predicted in the control horizon using current state and driver inputs. These control inputs are virtual but universal to any vehicles. Now feeding into the lower-level controller, the augmented torque or braking ($\Delta Q_1, \Delta Q_2, \ldots, \Delta Q_i$) or/and steering ($\Delta\delta_1, \Delta\delta_2, \ldots, \Delta\delta_i$) are optimally distributed by solving a constrained quadratic problem at each time step. Real time constraints bound the final control inputs to violate not, such as actuator limits, tire capacity. Not only, the lower-level control allocation provides optimal control actions, but also gives freedom and reconfigurability for different actuation systems.

### A. Prediction Model

The prediction model's accuracy and complexity have a critical impact on the closed-loop optimization, i.e. computational cost and performance. The accuracy of this paper is guaranteed using the nonlinear tire model meanwhile complexity is considered by linearizing the body dynamics and tire operating point. Regarding non-articulated vehicles, the prediction model is explicitly given in (6). But the final model of articulated vehicles is formulated in an indirect manner (14), which has difficulties to serve as a standard prediction model. A reformation is therefore introduced.

For a further simplification, assuming the articulation angle $\lambda$ is small, the kinematic relationships and force constraints of the articulation point is linearized to rewrite (14) as,

$$\begin{aligned} &\underset{2\text{ units}}{\text{s.s form.}} && \dot{x} = Ax + BF_{CG} + CF_H \\ &\text{s.t.} && M_{(2\times10)}\dot{x} = P_{(2\times10)}x \\ &&& N_{(6\times8)}F_H = Q_{(6\times10)} \end{aligned} \tag{16}$$

where,

$$M_{(2\times10)} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -l_h^t & 0 & 0 & 0 & -1 & -l_h^i & 0 & 0 \end{bmatrix},$$

and $$P_{(2\times10)} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -v_x^t & 0 & 0 & 0 & 0 & v_x^t & 0 & 0 \end{bmatrix}.$$

In the matrix $N_{(6\times8)}$, elements $N_{1,1}$, $N_{1,5}$, $N_{2,2}$, $N_{2,5}$, $N_{3,3}$, $N_{3,7}$, $N_{4,4}$, $N_{4,8}$, $N_{5,4}$, $N_{6,8}$ are equal to 1 and the rests are zeros. In the matrix $Q_{(6\times10)}$, elements $Q_{6,5}$ and $Q_{6,10}$ equal to $-K_{t-i}$ and $K_{t-i}$, respectively, and the rests are zeros. $F_H = [F_{x,h}^t\ F_{y,h}^t\ M_{z,h}^t\ M_{x,h}^t\ F_{x,h}^i\ F_{y,h}^i\ M_{z,h}^i\ M_{x,h}^i]^{\mathrm{T}}$ represents the forces at the hitch point. The first constraint in (16) means there are two independent linear algebraic equations in terms of $\dot{x}$ and $x$ while the second one means six independent linear algebraic equations in terms of ${F_H}^t$ and ${F_H}^i$. Multiply $M$ to both sides of ( 16), it gives,

$$M\dot{x} = MAx + MBF_{CG} + MCF_H \tag{17}$$

Combining ( 17) with the constraint $M\dot{x} = Px$, by taking the right-hand side of both equation, it gives,

$$\begin{aligned} &MAx + MBF_{CG} + MCF_H = Px \\ &\Rightarrow MCF_H = (P - MA)x - MBF_{CG} \end{aligned} \tag{18}$$

Next, integrating the constraint $N_{(6\times 8)}F_H = Q_{(6\times 10)}$ in one matrix operation form, it gives,

$$\left[\frac{(N_{6\times 8})}{(MC)_{(2\times 8)}}\right] F_H = \left[\frac{Q_{(6\times 10)}x}{((P-MA)x - MBF_{CG})_{(2\times 1)}}\right] \tag{19}$$

To achieve a form that connects the $F_H$ and state variables $x$ and $F_{CG}$, here a trick of matrix transform is used, it holds that,

$$\left[\frac{Q_{(6\times 10)}x}{((P-MA)x - MBF_{CG})_{(2\times 1)}}\right] = \left[\frac{Q_{(6\times 10)}}{(P-MA)}\right] x - \left[\frac{0_{(6\times 6)}}{MB}\right] F_{CG} \tag{20}$$

Since the matrix associated with $F_H$ holds full rank, $F_H$ can be expressed in terms of $x$ and $F_{CG}$,

$$F_H = \underbrace{\left[\frac{N_{(6\times 8)}}{(MC)_{(2\times 8)}}\right]^{-1}}_{J} \underbrace{\left[\frac{Q_{(6\times 10)}}{(P-MA)}\right]}_{K_1} x - \underbrace{\left[\frac{N_{(6\times 8)}}{(MC)_{(2\times 8)}}\right]^{-1}}_{J} \underbrace{\left[\frac{0_{(6\times 6)}}{MB}\right]}_{K_2} F_{CG} \tag{21}$$

Substituting it back to ( 16), the state-space formulations becomes,

$$\begin{aligned} \dot{x} &= Ax + BF_{CG} + C(JK_1x - JK_2F_{CG}) \\ \Rightarrow \dot{x} &= (A + CJK_1)x + (B - CJK_2)F_{CG} \end{aligned} \tag{22}$$

where $F_{CG} = [F_{CG}^{t\mathrm{T}} \quad F_{CG}^{i\mathrm{T}}]^{\mathrm{T}}$. Applying the same procedure where $F_{CG}$ is derived from a single vehicle unit, $F_{CG}$ can be expressed in terms of tire model along with vehicle states and driver inputs. Eventually, it will achieve the full model configuration similar with (6).

### B. High-Level MPC

The high-level MPC is designed to compute the corrective CG forces for stability's sake. MPC naturally handles multivariable systems without additional design complexity, thus simplifying the controller development with multivariables. Furthermore, states and actuator constraints are easily formulated into the optimization problem. Desired references of the lateral speed, the yaw rate, and articulation angle and the rollover index are generated and updated from the reference module,

$$y_d = \left[v_{x_d}^t \quad v_{y_d}^t \quad r_d^t \quad RI_d^t \quad \lambda_d \quad RI_d^i\right]^{\mathrm{T}} \tag{23}$$

Note that, (23) is the full state of a two-unit vehicle. When it comes to real implementation, the reference vector and prediction model are simplified as to meet the desired objectives. For instance, a yaw-plane dynamics control for tractor-trailer gives $y_d = [v_{y_d}^t \quad r_d^t \quad \lambda_d]^{\mathrm{T}}$. In an MPC scheme, the model of the plant is used to predict the future evolution of the system, and the controller action sequence is achieved by repeatedly solving finite time optimal control problems in a receding horizon fashion [30]. The optimization is formulated as a constrained quadratic program.

The linear-time-varying(LTV) model is discretized using the Euler method (Zero Order Holder) with sampling period $T_s$,

$$x(k+1) = A_d x(k) + B_d v(k) + G_d w(k) + d_d(k) \tag{24}$$

$$y(k) = C_d x(k) \tag{25}$$

where the current control vector $v(k) = [\Delta F_x^t \quad \Delta F_y^t \quad \Delta M_z^t \quad \Delta F_x^i \quad \Delta F_y^i \quad \Delta M_z^i]^{\mathrm{T}}$. $y(k)$ is the defined output used to track references. We define the cost function in a finte horizon $N_p$ as,

$$J(x(t), V_t) = \sum_{k=1}^{N_p} \left\|y_{t+k,t} - y_{d t+k,t}\right\|_Q^2 + \sum_{k=0}^{N_p-1} \|v_{t+k,t}\|_R^2 + \|v_{t+k,t} - v_{t+k-1,t}\|_S^2 \tag{26}$$

where $V_t = [v_{t,t}, \quad \cdots \quad v_{t+N_p-1,t}]^{\mathrm{T}} \in R^{6N_p \times 1}$, represents the optimization sequence at time $t$. Index $t+k, t$ denotes the predicted value at $k$ steps ahead of the current time $t$. The first term in (26) is the tracking error of the references. The second term is the prospective CG forces and minimizes the control efforts. The third term is optional and enforces proximity to the previous step to prevent oscillation in control actions. The positive semi-definite $Q$, $R$ and $S$ are weighting matrices that reflect the importance of these terms in the cost function. Alghouth not the focus of this paper, it is worthy to mention here, regarding vehicle stability objectives, they can be prioritized by regulating the corresponding weights of $Q$. Generally, priority from high to lower should be, rollover prevention > anti-excessive sideslip>yaw rate tracking. By making certain rules, one could achieve the objective prioritization [31]. At each time step the following finite horizon ($N_p$) optimal control problem is solved on-line,

$$\begin{aligned} \min_{V_t} \quad & J(x(t), V_t) \\ \text{s.t.} \quad & x(k+1) = A_d x(k) + B_d v(k) + G_d w(k) + d_d(k) \\ & y(k) = C_d x(k) \\ & y_{\min} \leq y_{k,t} \leq y_{\max} \\ & v_{\min} \leq v_{k,t} \leq v_{\max} \\ & v_{k,t} = v_{k-1,t} + \Delta v_{k,t} \\ & \left\|y_{t+N_p-1,t} - y_{d t+N_p-1,t}\right\|_Q^2 + \left\|u_{t+N_p-1,t}\right\|_R^2 \leq \gamma \\ & k = t, \ldots, t+N_p-1 \end{aligned} \tag{27}$$

*Iteration Algorithm:*

1. Acquire the new state $x(t)$, driver input $w(t)$, reference state $y_d(t)$;
2. Solve the QP optimization problem (27);
3. Apply $v(t) = v(t+0, t)$
4. $t \leftarrow t+1$. Go to 1.

*Remark:* The second last constraint in (27) refers to the terminal constraint to guarantee the stability the LTV MPC scheme. Detailed work could be found in [32]. The Batch approach is used to formulate the problem as a standard QP problem. The objective function can be expressed as a function of the initial state $x_{0,t}$ and sequences of reference, driver input. The solver, $qpOASES$, based on a structure-exploiting active-set method, provides fast and reliable solutions and is used in this research [33]. The corrective control vector should be bounded by the maximum CG forces/moment a specific vehicle can generate. The high-level controller hence completes the calculation of optimal corrective GG forces.

### C. Lower-Level Control Allocation

In this section, calculations by high-level MPC are applied through corner modular, i.e. drive/braking and steering. Articulated vehicles are usually over-actuated mechanical systems, i.e. brake system in more than two wheels; active steering system. The lower-level controller should be able to handle redundancy and various configuration of active actuators. We use the technique of Control Allocation (CA) to cope with such redundant actuation systems [17], [34]. As a result, CA offers an optimal distribution on actuator commands, meanwhile minimization of the CG corrective forces errors.

To achieve a dynamics stability, the requested corrective CG forces calculated by the high-level MPC are denoted by,

$$v^* = \begin{bmatrix} \Delta F_x^{t*} & \Delta F_y^{t*} & \Delta M_z^{t*} & \Delta F_x^{i*} & \Delta F_y^{i*} & \Delta M_z^{i*} \end{bmatrix}^{\mathrm{T}} \tag{28}$$

Using the mappings of aforementioned modeling, the actual CG forces generated by lower-level CA are denoted by,

$$v = \begin{bmatrix} \Delta F_x^{t} & \Delta F_y^{t} & \Delta M_z^{t} & \Delta F_x^{i} & \Delta F_y^{i} & \Delta M_z^{i} \end{bmatrix}^{\mathrm{T}} = B_p u \tag{29}$$

where $B_p = L_c T_c L_w T_w B_1$, is the control effectiveness matrix (or mapping matrix), that connects all the possible actuator configurations of the vehicle. That is why CA offers the great convenience to adapt different actuation configurations. $u$ is the lower-level control actions, e.g. torques or steerings defined in (5).

CA is formulated as a convex quadratic programming with linear constraints. Therefore, at each time step, the lower level controller solves the following optimization problem,

$$\begin{aligned} \min_{u} \quad & \xi \left\| v^* - B_p u \right\|_{W_e}^2 + \left\| u \right\|_{W_u}^2 \\ \text{s.t.} \quad & A_{eq} u = b_{eq} \\ & lb \le u \le ub \end{aligned} \tag{30}$$

where, the first term in the objective function of (30) is to minimize the CG forces error, while the second term is to minimize the control efforts of actuations. $lb, ub$ are the lower and upper bounds, $W_e, W_u$ are positive definite weighting matrices, which give a compromise between two costs. $\xi$ is a used to emphasize the importance of minimizing the allocation error, that typically set very large. The equality constraint ($A_{eq} u = b_{eq}$) could be useful while considering some engineering practices. For instance, active steering strategy usually makes the steering angles of the left wheel and right wheel equal. $W_e, W_u$ may be time-varying, depending on the driver inputs or road conditions. It reflects how the available actuators will be preferred and utilized for a integrated control strategy. This brings up a interesting topic, actuators prioritization, which is omitted for sake of brevity.

### D. Real-Time Constraints

This section discusses about the constraints from different sides. According to $u_i = \begin{bmatrix} \Delta Q_i & \Delta \delta_i \end{bmatrix}^{\mathrm{T}}$, $lb_i(2)/ub_i(2)$ denotes the lower/upper boundary of the active steering angles. They are mainly constrained by the maximum steering angle of the physical limitation. $lb_i(1)/ub_i(1)$ denotes the lower/upper boundary of the active torques (drive or brake),

$$lb_i(1) = \max(Q_i^{\text{brake}}, -Q_i^{\text{tire}}, -Q_i^{\text{fail}}) - Q_i \tag{31}$$

$$ub_i(1) = \min(Q_i^{\text{drive}}, Q_i^{\text{tire}}, Q_i^{\text{fail}}) - Q_i \tag{32}$$

where $Q_i$ is the estimated torque of wheel '$i$' from the driver's request. The lower/upper boundary is obtained by considering three sets of constraints, i.e. drive/brake torque capacity, tire force capacity and actuators failure.

$Q_i^{\text{drive}}$ is the maximum torque of the wheel that the powertrain can deliver, while $Q_i^{\text{brake}}$ is the maximum brake torque depending on the vehicle's braking system on wheels of different axles. $Q_i^{\text{tire}}$ is derived from a 'tire ellipse model' [35]. Given the estimated tire forces, i.e. $f_{xi}$, $f_{yi}$ and $f_{zi}$, the maximum 'available longitudinal force' is achieved. One may note that the wheel slip prevention is not included in this paper. However, it can be tackled by off-the-shelf modules, i.e., ABS and TCS. $Q_i^{\text{fail}}$ is the torque value when an actuator failure occurs. For example, a braking actuator locked in a faulty position. Control allocation has the ability to handle fault-tolerant control [34]. Thanks to updating constraints in real time, once a fault is detected, CA will systematically set the lower and upper boundaries to the locked value or zeros assuming a complete failure. There is no need to redesign the control laws as the system can allocate the remaining actuators automatically as to best meet the control tasks.

## V. Applications

### A. Matrix-Size Reduction

MPC needs larger computational load and memory caches than classical controls. There are challenges while implemented within auto-grade electronic control units (ECUs). In addition to selecting a fast and robust QP solver, another key issue is the dimension of the system matrices and the number of control inputs. While applying the proposed controller to a specific case, in practice, the configuration of the vehicle to be controlled is prior knowledge, like, axle/articulation configuration; drive mode and steering system; control objective; active control system. Therefore, it is necessary to adjust the matrix dimension for computation's sake. The controller algorithm flowchart in Fig. 4 suggests how it works while giving a vehicle for dynamics control. In offline mode, once the target vehicle is chosen, the vehicle configurations, stability objectives and physical constraints are surely acquired. Therefore, 'Boolean Matrices,' namely, $T_\varepsilon, T_c$ and $T_w$ are defined. Dimensions of model matrices($A_d$, $B_d$, $G_d$, $d_d$ and $C_d$), the mapping matrix ($B_p$), and vectors ($x$, $y$, $v$ and

TABLE II
APPLICATIONS DESCRIPTION

| App. | Vehicle | Driver and Maneuver | Road | Control Objective | Active Actuator |
|---|---|---|---|---|---|
| I | Articulated bus* | Sine steer(one cycle impulse), with constant target speed of 80 km/h, | $\mu = 0.75$ | -Trailer sway control<br>-Yaw tracking control | Active trailer steering(ATS) |
| II | Articulated bus† | Sine steer(one cycle impulse), with initial speed of 85 km/h, no throttle | $\mu = 0.75$ | -Trailer sway control<br>-Yaw trackability of tractor | Differential braking system(DBS) (multi configurations) |
| III | Articulated truck+ | 1) Double lane change, initial speed of 80 km/h<br>2) Brake in a step turn, initial speed of 70 km/h | 1) $\mu = 0.75$<br>2) $\mu = 0.5$ | -Rollover prevention<br>-Jackknifing prevention | Tractor differential braking |

* Articulated bus: has three axles (axle #1& 4 of unit 1, and axle #4 of unit 2), with 2nd axle drive powered by pure battery; with active trailer steering ($-8^\circ$ to $8^\circ$), shown in Fig. 5.
† Articulated bus: has three axles (axle #1& 4 of unit 1, and axle #4 of unit 2), with 2nd axle drive powered by pure battery; with all wheels differential braking, shown in Fig. 7.
+ Articulated truck: has four axles(axle #1,2& 4 of unit 1, and axle #3 &4 of unit 2), with tractor rear axles drive from an IC engine; with tractor differential braking,shown in Fig. 9.

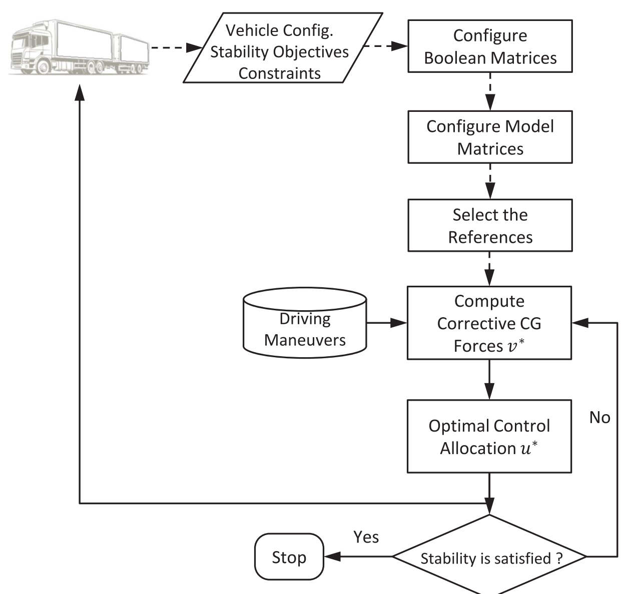


Fig. 4. Controller algorithm flowchart.

$u$) are simplified and determined. All matrices are configured to be the proper size before execution. In online mode, it follows the two-layer control framework as introduced above. There are three applications listed in Table II to demonstrate as well as verify the proposed framework.

### B. Controller Parameters and Tuning

There are a great number of research can be found on MPC tuning methods and industrial applications, where an excellent review is elaborated in [36] and a detailed study is presented in [37]. Generally, there are three significant goals when tuning a MPC controller [38]: (1) achieving appropriate model predictions over the horizon; (2) a compromise between robustness and performance; (3) a feasible computational cost.

*Sample Time & Prediction Horizon:* The controller sampling time $T_s$ is for use in the discrete-time formulation to predict the future states.If $T_s$ is too big, the model might lose accuracy, and when a disturbance comes in, the controller will not be able to react to disturbance fast enough. On the contrary, if the sample time is too small, the controller can react much faster and setpoint changes, but costs high computations. Therefore, there is a trade-off. In order to have accurate predictions, $T_s$ should be small enough that allows 3-10 steps in the settling of a fastest dynamics [39]. Others recommend that the sampling time is between 10% to 25% of the system rise time [40]. By studying the open-loop response of the vehicle system, this research tentatively chose 0.015/0.02 s and calculated that $N_p = 10$ for controller design.

*Control Horizon:* The future control actions lead to the predicted future output, the number of control moves to time step $N_c$ are called the control horizon. The smaller the control horizon, the fewer the computations. However, the minimum horizon will not give us the best control maneuvers. This leads to a trade-off. If the control horizon is increased, it results in a more robust but more aggressive controller as well as larger computational cost. On the contrary, decreasing the control horizon creates more the conservativeness and saves computations but with the price of reducing robustness [36]. It is ideal to set $N_c$ large enough to cover all significant adjustments of the control sequence in order to cope with a set-point change. Georgiou *et al.* [41] chose the time that takes for the output response to reach 60% of steady state as control horizon. This research assumes the controller horizon is equal to prediction horizon for simplicity sake.

*Weights:* There are weights on the outputs and weights on the control efforts for both high-level and lower-level controllers. Weights tuning depends on system requirements and control desires. For instance, if one emphasizes the tracking accuracy and wants a more aggressive control performance, more weights should be put on the tracking error side, i.e., weighting matrix $Q$. If robustness and smooth control performance and energy saving are important, one could increase the weighting matrix $R$ to realize such goal. In this research, a normalized tuning method is used for weight tuning [42]. To achieve a robust and satisfying performance, Trial and error is necessary to see the changes and balance of the control responses. In addition, when it comes to production implementation, a fine tuning process is needed that considers all possible situations, such as, road slope, bank angle, uncertainties from road friction and side wind. Taking the Application II (articulated bus with DBS) as a example, Table III listed the parameters of the controller. One could follow the similar process to tune the controller for other applications.

### C. Application I: An Articulated Bus With ATS

In this case, an articulated bus is built in TruckSim by modifying the templates of it. Trailer sway of tractor semi-trailer is a serious threat for vehicle safety. In this case, the controller

TABLE III
CONTROLLER PARAMETERS OF APP.II

| Symbol | Description | Value |
|---|---|---|
| $T_s$ | Controller sample time (s) | 0.015 |
| $N_p$ | Steps of the prediction horizon | 10 |
| $N_c$ | Steps of the control horizon | 10 |
| $\alpha^{sat}$ | Tire saturated side slip angle (°) | 7.5 |
| $q_1$ | Weight of lateral speed tracking error | 32.4 |
| $q_2$ | Weight of yaw rate tracking error | 1800 |
| $q_3$ | Weight of art. angle tracking error | 3600 |
| $r_1$ | Weight of tractor yaw moment | 5e-09 |
| $r_2$ | Weight of trailer yaw moment | 2e-08 |
| $\Delta M^t_{z,min}/\Delta M^t_{z,max}$ | Bounds of tractor yaw moment (Nm) | -40000/40000 |
| $\Delta M^i_{z,min}/\Delta M^i_{z,max}$ | Bounds of trailer yaw moment (Nm) | -20000/20000 |
| $W_e$ | Weight of CG forces mapping error | diag([50,50]) |
| $W_u$ | Weight of control efforts | 0.02∗diag([1,1,1,1,2,2]) |
| $Q^{brake}$ | Tractor/trailer brake limits (Nm) | -8000/-8000 |

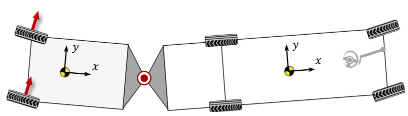


Fig. 5. A articulated bus with active trailer steering.

aims to tackle the problem. With the prior-knowledge of the configuration in Table II and Fig. 5, the 'Articulation, Axle and Actuator Boolean Matrix' can be written as,

$$T_\varepsilon = I_{10\times 10};$$
$$T^t_c = diag(1,1,1,1,0,0,0,0,0,0,0,0,1,1,1,1);$$
$$T^i_c = diag(0,0,0,0,0,0,0,0,0,0,0,0,1,1,1,1);$$
$$T^t_w = diag(0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0);$$
$$T^i_w = diag(0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,1).$$

Since the vehicle configuration is given and control objectives are targeted, matrix dimensions of state space model (24) and (25) are specified: $A_d \in R^{4\times 4}$; $B_d \in R^{4\times 2}$; $G_d \in R^{4\times 4}$; $d_d \in R^{4\times 1}$; $x = [v^t_y \quad r^t \quad v^i_y \quad r^i]^{\mathrm{T}}$; $v = \Delta M^i_z$; $y = [v^t_y \quad r^t \quad \lambda]^{\mathrm{T}}$. Regarding the mapping equation (29) and control input, where $B_p \in R^{1\times 2}$; $u = [\Delta\delta_1 \quad \Delta\delta_2]^{\mathrm{T}}$ and $\Delta\delta_1 = \Delta\delta_2$.

A sharp sine impulse steer, with the amplitude of 75 deg and period of 2 s, described in and top-left corner of Fig. 6 is performed for this simulation. A constant speed is hold for both OFF and ON modes for fair comparison purpose. In the control 'OFF' mode, a severe trailer sway and tractor yaw oscillation are triggered. The articulation angle reached as big as 10 deg and and damped very slow. Note that, the maximum articulation angle is translated to be around 3 m of the arc length the rear end swept, which is very dangerous due to the violation to its driving lane. In addition, neither the leading unit(tractor) is stable in yaw motion. Through active trailer steering, the articulation angle is well tracked throughout the maneuver and the sway stopped after 5th second. As the secondary objective, the tractor yaw rate tracking shows similar evidence that the vehicle is stabilized. From the result of active steering angle, it is notable that small steering adjustments are only needed but the stabilization impact is huge.

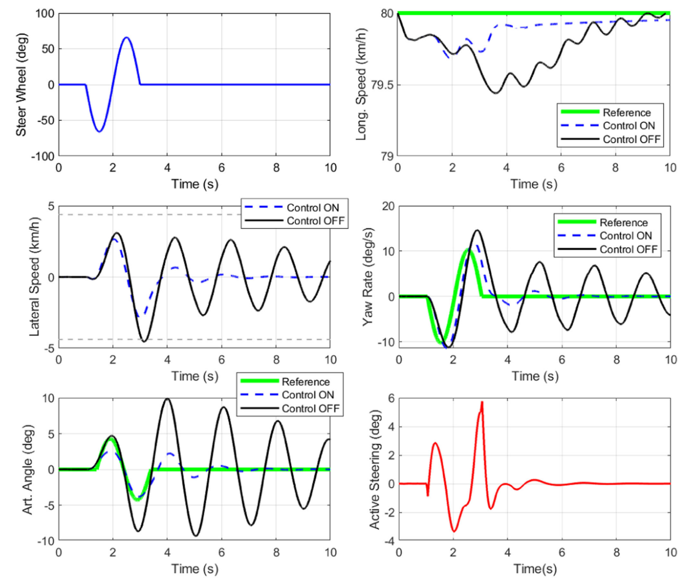


Fig. 6. Results of yaw and sway control of App. I.

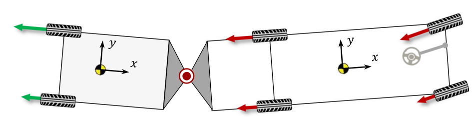


Fig. 7. A articulated bus with differential braking.

## D. Application II: An Articulated Bus With DBS

In this case, the same articulated bus in App.I but configured with differential braking is built in TruckSim. The main purpose of this application is to compare the performance of various differential brake systems, as well as demonstrate the reconfigurabitliy of the framework. With the prior-knowledge of the configuration in Table II and Fig. 7, the 'Articulation and Axle Boolean Matrix' are kept same as App. I, but 'Actuator Boolean Matrix' is modified as,

$$T^t_w = diag(1,0,1,0,0,0,0,0,0,0,0,0,1,0,1,0);$$
$$T^i_w = diag(0,0,0,0,0,0,0,0,0,0,0,0,1,0,1,0).$$

Given the vehicle configuration and control objectives, matrix dimensions of state space model (24) and (25) are specified: $A_d \in R^{4\times 4}$; $B_d \in R^{4\times 2}$; $G_d \in R^{4\times 4}$; $d_d \in R^{4\times 1}$; $x = [v^t_y \quad r^t \quad v^i_y \quad r^i]^{\mathrm{T}}$; $v = [\Delta M^t_z \quad \Delta M^i_z]^{\mathrm{T}}$; $y = [v^t_y \quad r^t \quad \lambda]^{\mathrm{T}}$.

For the mapping equation (29), there are three versions of the control strategies listed and named in Table IV. For 'Controller A' where $B_p \in R^{1\times 4}$; $u = [\Delta Q_1 \quad \Delta Q_2 \quad \Delta Q_3 \quad \Delta Q_4]^{\mathrm{T}}$; 'Controller B' where $B_p \in R^{1\times 2}$; $u = [\Delta Q_5 \quad \Delta Q_6]^{\mathrm{T}}$; 'Controller C' where $B_p \in R^{2\times 6}$; $u = [\Delta Q_1 \quad \Delta Q_2 \quad \Delta Q_3 \quad \Delta Q_4 \quad \Delta Q_5 \quad \Delta Q_6]^{\mathrm{T}}$; Thanks to the reconfigurability of the proposed framework, we can easily design the different controllers and compare the performance.

Similar to App. I, a sharp sine steer, with the amplitude of 75 deg and period of 2 s ( Table II), is applied to the bus. In the control 'OFF' mode, the bus experienced severe trailer sway, the articulation angle reached as big as 10 deg, and the vehicle gets stable gradually. The tractor yaw rate oscillated

TABLE IV
PERFORMANCE OF DIFFERENT CONTROLLER OF APP. II

| Controller | Active Actuators | Max. $\lambda$ | Duration of Sway | Art. Angle Tracking | Yaw Rate Tracking |
|---|---|---|---|---|---|
| OFF | No active actuators | 10 | $1^{st} - 11^{th}$ | Trailer sways | Oversteer |
| A | Differential braking on tractor | 8.5 | $1^{st} - 6.5^{th}$ | Limited | Effective |
| B | Differential braking on trailer | 6 | $1^{st} - 6.5^{th}$ | Reduced with a delay | Reduced with a delay |
| C | Differential braking on tractor& trailer | 6 | $1^{st} - 6.5^{th}$ | Effective | Effective |

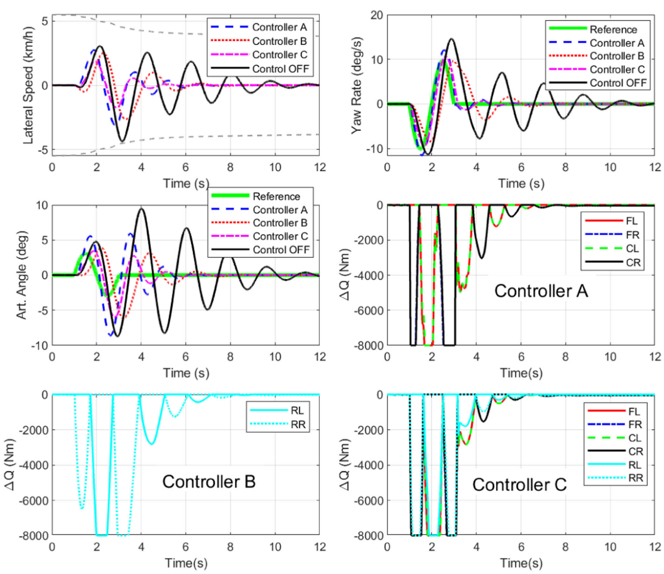


Fig. 8. Trailer sway control performance comparison using different active actuator systems of App. II.

in the same way and time range. By engaging active control systems, namely, Controller A, B, C, the stabilization is achieved in a varying extent, described in Table IV. The trailer sway is effectively restrained in terms of maximum articulation angle and sway duration. Interestingly, the comparison finds that, Controller A is more effective on yaw rate tracking control while the Controller B is more effective on sway stabilization but with a delay. This makes sense intuitively, as Controller A uses the differential braking of the tractor while Controller B uses that of trailer. Besides, Controller C shows the very good comprehensive performance on both control objectives.

### *E. Application III: A Commercial Truck*

Many tractor trailer combinations have the changeable trailer. However, the scope of this work assumes the trailer configuration, e.g. axles, is constant and its information is known, measured/estimated. In this case, a commercial truck with full payload is built in TruckSim. As mentioned, truck dynamics is complex and challenging for active controls. In this case, the controller aims to prevent the likelihood of trailer rollover and jackknifing, which are the most typical threat to highway safety of long trucks. With the prior knowledge of the configuration in Table II and Fig. 9, the 'Articulation, Axle and Actuator Boolean

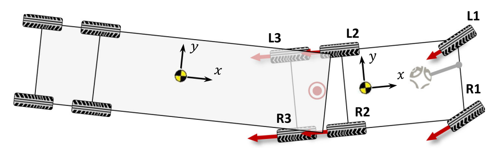


Fig. 9. A articulated truck configuration.

Matrix' can be written as,

$$T_\varepsilon = I_{10\times10};$$
$$T_c^t = diag(1,1,1,1,0,0,0,0,1,1,1,1,1,1,1,1);$$
$$T_c^i = diag(0,0,0,0,0,0,0,0,1,1,1,1,1,1,1,1);$$
$$T_w^t = diag(1,0,1,0,0,0,0,0,1,0,1,0,1,0,1,0);$$
$$T_w^i = diag(0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0).$$

Again, with the knowledge of the vehicle configuration and control objectives, matrix dimensions of state space model (24) and (25) are specified: $A_d \in R^{8\times8}$; $B_d \in R^{8\times1}$; $G_d \in R^{8\times6}$; $d_d \in R^{8\times1}$; $x = [v_y^t \quad r^t \quad \phi^t \quad \dot{\phi}^t \quad v_y^i \quad r^i \quad \phi^i \quad \dot{\phi}^i]^{\mathrm{T}}$; $v = \Delta M_z^t$; $y = [r^t \quad \lambda \quad RI^i]^{\mathrm{T}}$.

For the mapping equation (29), where $B_p \in R^{1\times6}$; $u = [\Delta Q_1 \quad \Delta Q_2 \quad \Delta Q_3 \quad \Delta Q_4 \quad \Delta Q_5 \quad \Delta Q_6]^{\mathrm{T}}$; Note that, in this case, the high-level controller only generates the yaw moment of tractor and the lower-level controller computes the active brake commands, where, from left to right and front to back, are indicated by L1, R1 to L3, R3 in Figs. 9 and 10.

In the maneuver of double lane change (DLC) in Fig. 10, a driver is in the loop using a path-following model of TruckSim. The driver performed a harsh DLC to avoid potential obstacles on a highway. The weights of $Q$ are set to emphasize rollover prevention. The rollover threshold ($RI_{\max}$) is set to be 0.5 (see the 'gray dashed line'). If no active control engaged, the trailer rollover likelihood is seen at 5th-6th, and 7.5th second. When differential braking is introduced, the maximum load transfer ratio of the trailer is reduced by around 36.5% from $-0.82$ to $-0.52$. In addition, the lateral acceleration shows similar evidence on rollover prevention. Although the weights of yaw rate and articulation angle (in $Q$) are set very small, the tractor yaw rate and articulation are also reduced because of the corrective yaw moment and speed drop.

In the second case, to trigger a jackknife, a step steer and a step brake of 15 MPa at 2nd second are applied. The tires are in a high slip or saturation scenarios at the low road friction coefficient condition. And the huge inertia of the trailer due to the full load contributes to the likelihood of jackknife. As

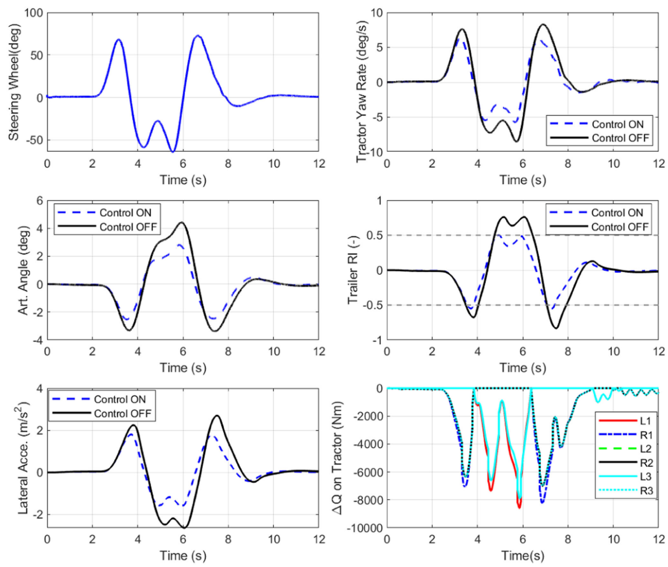


Fig. 10. Results of rollover prevention of App. III.

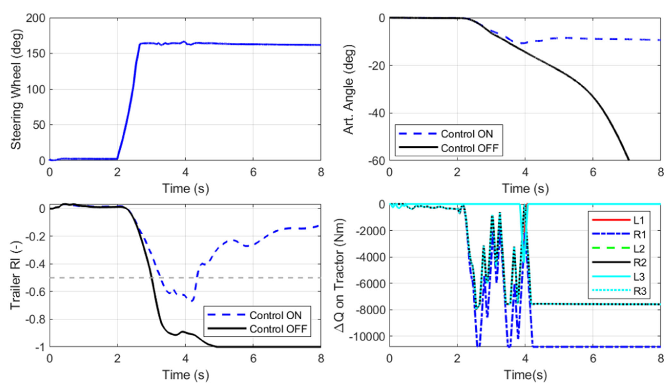


Fig. 11. Results of anti-jackknifing of App. III.

shown in Fig. 11, in Control OFF Mode, the articulation angle kept increasing and diverged to around 90° at the 8th second. Besides, the trailer RI stayed in the high value during the turn. However, the engagement of differential braking prevented the truck from jackknife successfully, where the articulation angle is reduced to 10° and RI is dropped to a relatively safe area in practice.

## VI. Conclusion

Knowing that articulated/multi-axle vehicles come with significantly various configurations, this paper presented a universal and reconfigurable control methodology that applies to any vehicles dynamics control. The integrated control formulation covers longitudinal, yaw, lateral and roll stability objectives and one can control any combinations of them depending on the vehicle's configurations and needs. Moving from one vehicle to another does not require re-designing the control system, and major re-tuning is also avoided. We first introduced a reconfigurable modeling and extended to articulated vehicles, and then a two-layer control system is designed to cover the diversity of vehicle configurations, multi-input multi-output, over-actuation, fault tolerance problems. Two levels are formulated as optimization-based control problems wphen potential constraints are considered. In the light of three diverse applications presented, it is shown that the proposed system is universal and reconfigurable as well as effective on the vehicles' stability control.

In future work, actuators prioritization will be interesting to investigate. Suppose the vehicle has more than two active actuation system, e.g., differential braking system and active steering control. Building certain rules on actuators prioritization and tuning the controller weighting matrices in realtime is a potential to achieve so. In addition, it is planned to implement the control system in the experimental tractor-trailer, so as to validate the effectiveness and reconfigurability.


## Acknowledgment

The authors would like to acknowledge the financial support of the Ontario Research Fund (OTF) and Natural Sciences and Engineering Research Council (NSERC) of Canada in this work.


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